

Transfer Optimization for Heterogeneous Drone Delivery and Pickup Problem

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Abstract—Recently, with the enhancement of drone’s payload and endurance capability, drone logistics has received extensive attention from giant logistics enterprises. This research delves into the Heterogeneous multi-Drone Delivery Pickup Problem (HD-DPP) through a novel distribution mode. Here, a large drone ferries multiple smaller drones to a specified sub-region. These smaller drones oversee parcel delivery and pickup tasks before returning to automated airports for recycling. Then we formulate it into an open two-echelon multi-objective routing optimization problem, addressing pickup and delivery problems. To enhance optimization efficiency, relying on conventional optimization algorithms initiates the optimization process from an initial state, neglecting pertinent empirical knowledge embedded in optimized instances. This oversight results in a considerable squandering of computational resources. Based on the above motivation, we propose a Transfer Optimization with Heuristic Operators (TOHO) method. TOHO adopts insights from previously optimized task models, ensuring quick convergence for new tasks. We also present a mapping method rooted in topological alignment, enabling effective alignment between source and target instances, which fosters faster optimization convergence. Furthermore, we design a voting-mechanism-based ensemble genetic optimization strategy employing heuristic operators to select the Pareto frontier with high-quality diversity and convergence. Extensive experimental results on extended benchmark instances validate the efficacy of TOHO, and underscore its capability to speed up the optimization while ensuring superior knowledge transfer compared to baselines.

Index Terms—Transfer optimization, combinatorial optimization, heterogeneous multi-drones, routing.

I. INTRODUCTION

AS THE demand for time-efficient delivery of goods is growing, such as take-away or fresh food, and other daily necessities, reliance solely on ground vehicles has proven insufficient for the needs of modern logistics. In contrast, drone technology offers notable advantages in energy conservation, adaptability, and punctuality, particularly for delivering lightweight packages with high added-value [1]. Given the advancement of

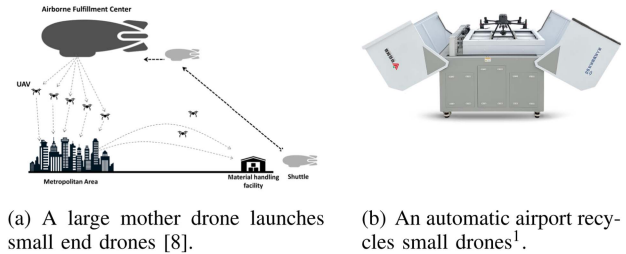


Fig. 1. Heterogeneous multi-drone delivery scenario.

drone technology, its use for deliveries has captivated not only industry behemoths like Amazon, DHL, Google, Meituan, and Jingdong, but also the research community [2], [3], [4]. Both Amazon and SF Express are pioneering a three-echelon logistics distribution model consisting of “trunk ultra-large drone + branch large drone + end small drone”. The branch drone carries several end drones released into the distribution region to enhance their operational duration. The end drones, being small and rechargeable, handle the delivery of lightweight parcels to customers. This delivery schema was first unveiled by Amazon in their Air Travel Center patent, employing drone distribution, and was put into action in 2013 [4]. The United Parcel Delivery Service (UPS) projects that replacing trucks with drones for last-mile delivery could yield an annual savings of roughly \$50 million [5]. Notably, by August 2021, Google’s Wing drone had successfully executed over 100,000 parcel deliveries [6]. In our earlier studies, we presented an innovative logistics distribution approach named heterogeneous multi-drone delivery [7]. This approach employs a large mother drone carrying several smaller end drones and their packages. Originating from a depot or distribution center, the mother drone traverses through K sub-regions, as depicted in Fig. 1(a). In each sub-region, a small end drone undertakes multiple deliveries to the designated customers. Once delivery is concluded, the end drone proceeds to an automated airport for recycling, as illustrated in Fig. 1(b).

Though our delivery model has shown great potential in future distribution systems, it considers the scenario where delivery exists only. In real-world parcel distribution scenarios, it is not only to deliver the customer’s parcel but also to collect the customer’s parcel, including return or exchange services. Thus, this paper focuses on the study of heterogeneous drone delivery and pick-up problem (HDDPP). Essentially, the

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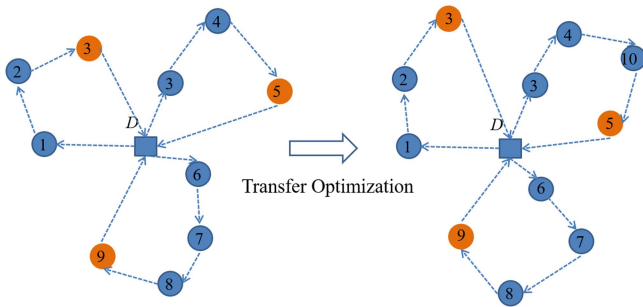


Fig. 2. Transfer optimization approach: target task VRP learned the prior knowledge from a source-optimized task.

HDDPP is extended from the two-echelon routing problem (2E-VRP). However, when facing a new optimization instance, the traditional optimization algorithms start exploring from scratch without applying past optimization experience. In fact, in real optimization scenarios, several solved optimization problems or instances may be similar to the current optimization problem. By utilizing the optimization experience of these similar problems or instances, the efficiency of optimizing the current optimization problem may be greatly improved. Unfortunately, existing optimization algorithms show difficulties in leveraging past optimization experience.

Recently, transfer optimization has gained much attention for its ability to transfer certain cultural and generalization abilities of humans [9], [10], [11], [12]. Transfer optimization utilizes prior knowledge from optimized related tasks and achieves performance acceleration [13]. However, the attention given to knowledge reuse based on transfer optimization has been far from sufficient. In addition, while transfer optimization has the potential to improve convergence speed, the actual transfer may be hindered by harmful negative transfer or hinder positive transfer [14], [15]. A key factor in achieving positive transfer is the degree of similarity between the search spaces of the source task and target task, i.e., whether the dimensions of the search spaces of the source task and target task are similar and whether the optimal solutions in these spaces have significant overlap [15]. If the overlap is high, learning knowledge from the source task will lead to positive transfer and accelerate the convergence speed of the target task optimization [16]. However, to the best of our knowledge, no such strategies have been designed to address HDDPP.

Hence, to better solve HDDPP, the present study aims to address the gap by drawing on past experiences and knowledge from similar optimized problems. To achieve this, we investigate the transfer optimization method with heuristic operators. Accordingly, we propose a topological alignment strategy to induce positive transfer for heterogeneous multi-drone delivery tasks, thus accelerating optimization convergence.

For instance, in Fig. 2, the best VRP routing solution is obtained in the source optimized task, and the corresponding topology of the source task is learned in the target task VRP. Specifically, the target task learned multiple vehicle routes from the source task, with routes 1 and 2 being completely learned. This emphasizes the importance of transfer optimization, which

activates useful knowledge and prevents the negative transfer of harmful knowledge. By learning the search space mapping function, we can induce the transfer of positively correlated knowledge to the target task in the multi-objective space of different optimization tasks, thereby improving the optimization performance in the transfer optimizer. However, in some cases, such as in route 3, where the 10-th customer node needed to be serviced but did not learn in the transfer optimization, heuristic strategies tailored to the problem characteristics are still required for optimization, despite obtaining high-quality solutions using prior knowledge.

Contributions: The contributions of this study are summarized as follows:

- A novel mode is proposed to address the heterogeneous multi-drone pickup and delivery logistics problem, namely HDDPP. Specifically, a large drone is utilized to transport multiple small drones to a designated sub-region. The small drones are responsible for delivering or picking up multiple parcels in a flight and finally fly to automatic airports for recycling. As such, a heterogeneous multi-drone routing mathematical model considering pickup and delivery is formulated.
- A transfer optimization with heuristic operators and strategies, namely TOHO, is proposed to speed the optimization process of HDDPP, where a mixture model consisting of multiple prior knowledge blocks is constructed. This approach combines both transfer optimization mechanisms and heuristic mechanisms. As a result, the convergence rate of similar target instances is significantly accelerated.
- In the transfer optimization mechanism, a probabilistic mixture model is designed, and a topological alignment is proposed to facilitate the transfer of knowledge. By doing so, we achieve the positive transfer of knowledge from the source model to the target instance, thus avoiding the need for a gradual search from a basic starting point.
- In the heuristic mechanism, an ensemble genetic optimization strategy with heuristic operators is designed, in which several specialized mutation and crossover operators are designed that are tailored to the problem domain. Meanwhile, a voting-based ensemble strategy is proposed to effectively select the Pareto frontier with high-quality diversity and convergence.
- Extensive experiments have been conducted on three large-scale extended benchmark instances and a real-world instance for HDDPP, and the results show that the proposed TOHO method significantly speeds up the convergence than the other three baselines.

The remainder of the paper is organized as follows. In Section II, recent transfer optimizer methods are briefly reviewed. In Section III, a new open two-echelon multi-objective routing model considering simultaneous pickup and delivery for heterogeneous multi-drone logistics is built. In Section IV, a mixture model-based transfer optimizer with heuristic operators is proposed. Section V reports and discusses the experimental results. Section VI covers the conclusion and future research directions.

II. RELATED WORKS

Transfer optimization and transfer learning, leveraging knowledge from analogous source tasks to facilitate the optimization of novel or target tasks, has garnered substantial attention and success across a diverse spectrum of applications [17], [18], [19]. This methodology has demonstrated efficacy not only in accelerating the learning process but also in augmenting the generalization performance of the targeted predictive model [20], [21]. In contrast, traditional heuristic optimization algorithms typically start optimization from scratch, without exploiting the available knowledge from previously optimized tasks or recognizing similarities between the current problem instance and past optimized instances [13]. In the initial evolutionary transfer optimization, Gupta et al. [22] proposed the multi-meme evolutionary algorithm (MFEA-I), which incorporates the meme algorithm framework into multitasking optimization, and introduced an innovative evolutionary multitasking optimization paradigm. Based on MFEA-I and the concept of online transfer optimization, Bali et al. [23] developed MFEA-II and designed an online knowledge transfer optimization algorithm based on a mixture probability model. Then they proposed MO-MFEA-II, a multitasking transfer optimization paradigm with multiple objectives [24]. Essentially, both multitask learning and Transfer Optimization involves acquiring optimized knowledge from prior experiences (or from the instance currently undergoing optimization) during the optimization process, facilitating the transfer of forward knowledge to expedite the convergence of the current task or instance.

Recently, beyond direct genetic transfer from source to target evolutionary searches, higher-order model-based transfer algorithms have emerged [25]. Kaedi et al. [26] employed a heuristic criterion within a specific class of combinatorial problems to retrieve related source probabilistic models from a database, guiding the target optimization search. Feng et al. [27], [28] introduced a positive semi-definite matrix in the model, inducing a biased distance metric for graph-based clustering cum sequencing problems. More recently, Feng et al. [29] proposed learning a mapping from a continuous source domain to the target search space through a single-layer denoising autoencoder. Iqbal et al. [30], [31] demonstrated the reuse of knowledge building blocks in the form of code fragments extracted from small-scale problems while addressing more complex, large-scale problems. Feng et al. [27], [28] adopted the form of a semi-positive definite matrix in the transfer optimization model, which introduced a biased distance measurement algorithm for graph-based clustering and ranking problems. They proposed to learn the mapping relationship from the search space of the source task to the target search space by introducing a single-layer denoising auto-encoder [29].

However, existing literature on expediting optimization through knowledge transfer in the context of vehicle routing is limited. Gupta et al. [32] provided an extensive overview of transfer optimization application scenarios. Feng et al. [27] proposed an alternative approach. Rather than directly reusing past optimized routing solutions, they advocate for the learning of a transformation matrix from such solutions. This matrix is

then utilized to guide the initialization of routing solutions for the evolutionary search when encountering new routing problems. Cunningham et al. [33] stood out for their proposition involving the direct storage and reuse of established high-quality solutions from previous problem instances in solving traveling salesman problems. This approach may encounter limitations when applied to problems that exhibit disparities in structural properties, such as variations in the number of customers, and topological structures. Feng et al. [34] applied multitasking optimization to combinatorial optimization problems and proposed an explicit multitasking optimization paradigm for vehicle routing problems.

Although transfer optimization holds the potential to improve the convergence rate, it can be affected by harmful negative transfer or hinder positive transfer, which may lead to the failure of transfer optimization [14], [15]. To mitigate the risk of incorporating detrimental knowledge, several methods seek to infer the synergy between problems offline [27], [28]. Researchers have explored ways to strengthen positive transfer between different optimization problems [29], [35], [36]. Feng et al. [29] proposed an automatic coding evolutionary search paradigm with the cross-heterogeneous problem, which learns the linear transformation of solutions in the continuous search space and transfers the optimized solutions to a series of multi-objective benchmark optimization problems. On the other hand, inhibiting harmful negative transfer is also a significant research problem. Researchers have proposed evolutionary transfer optimization methods specifically designed to inhibit negative transfer [37], [38], [39]. Da et al. [37] proposed a model-based adaptive transfer evolutionary algorithm (AMTEA), which can suppress the negative transfer of knowledge from tasks from different sources and presented experimental verification in combinatorial optimization problems. AMTEA captures the source-target similarity through the optimal stack of probabilistic search distribution models and uses this similarity to determine the degree of knowledge transfer between tasks [40]. However, the effectiveness of AMTEA largely depends on the similarity of the search space between the source task and the target task.

Based on the aforementioned related works review, it is discernible that while transfer optimization algorithms can leverage valuable knowledge from prior experiences to expedite convergence rates, they are susceptible to the phenomenon of negative knowledge transfer in combinatorial optimization problems. Based on this motivation, this paper aims to use the optimized experience and knowledge from similar problems in the past to study the transfer optimization algorithm to accelerate the optimization speed of target tasks, explore the adaptive strategy, and study the topological alignment strategy of the vehicle routing problem in the search space to obtain more knowledge positive transfer.

III. PROBLEM DESCRIPTION AND MODEL DEVELOPMENT

In this section, we discuss the modeling of HDDPP. The aim of HDDPP is to decrease distribution costs, and promptly respond to customer needs while using a reduced number of small drones. In the HDDPP, the large drone supports the small drones in

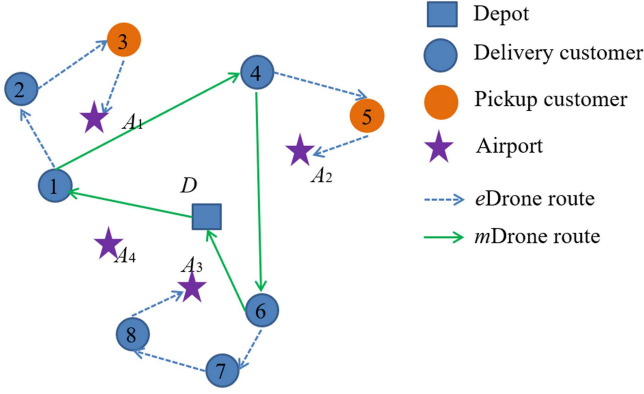


Fig. 3. The diagram of HDDPP routing problem.

logistics distribution by transporting them to sub-regions for takeoff, before returning to the distribution center. The small drones deliver and pickup parcels in sub-regions, and ultimately fly to an automatic airport for recycling. The automatic airport has the capacity to recycle multiple small drones and store a limited number of parcels. As illustrated in Fig. 3, drone 1 delivers parcels to customer nodes 1 and 2, then proceeds to customer node 3 to pickup parcels, and finally carries the parcels flying to the automatic airport A_1 for recycling. To comprehensively assess this problem, two objective functions are considered: the distance cost, and the number of small drones.

A. Problem Descriptions and Assumptions

The routing problem of HDDPP is defined as follows: The heterogeneous multi-drones consist of one large drone and K small drones, which serve n customer nodes. Among these customer nodes, m nodes are delivery nodes (shown as blue nodes in Fig. 3) and $n-m$ nodes are pickup nodes (shown as purple nodes in Fig. 3). The large drone departs from the depot D carrying K small drones and their parcels, and flies to K sub-regions to launch the small drones one by one. After launching one small drone in each sub-region, the large drone returns to the depot. Meanwhile, the small drone delivers and picks up parcels in a sub-region. Finally, the small drone flies to a nearby automatic airport for recycling. Fig. 3 presents the diagram of HDDPP routing problem.

Based on the above problem definition, the following model assumptions are provided to clarify the research scope of the HDDPP:

- 1) The distribution system comprises one large drone, and K small drones of the same type. The large drone is endowed with sufficient payload and range capabilities, while the small drones have a payload capacity of Q_S .
- 2) Both delivery customer nodes (C_D) and pick-up customer nodes (C_P) are taken into consideration. The customer node has only one attribute, i.e., pick-up or delivery.
- 3) The large drone is not responsible for delivering or picking up parcels directly. Instead, it transports small drones to sub-regions and launches them one by one. The small drones are responsible for parcel delivery and pickup in sub-regions.

- 4) The large drone flies K sub-regions sequentially, launches a small drone in each sub-region, and finally returns to the depot for recycling.
- 5) The large drone launches a small drone over a customer node in a sub-region instead of the clustering center.
- 6) Each small drone delivers and/or picks up multiple parcels, with the total weight of the parcels carried not exceeding its maximum payload, and finally flies to the automatic airport node for recycling.
- 7) Each customer node can only be serviced once by a small drone.
- 8) Multiple objectives are taken into consideration for simultaneous optimization, i.e., the number of small drones, and distance cost.

B. Symbols and Definitions

We define a large drone as the mother drone and the small drones as the end drones according to [41]. To accurately express the model, we define the following symbols, including the sets, parameters, and decision variables of the mother drone and end drones. The definition of each symbol is given in detail in Table I.

C. Mathematical Formulation

In the proposed HDDPP, for the purpose of clarity and conciseness, we define C_d as the node set of customers to be delivered and C_p as the node set of customers for pick-up. It should be noted that $C_d \cup C_p = C$. The delivery and pick-up services are carried out by end drones, while the mother drone is not directly involved in these services. Its role is limited to transporting the end drones to the designated sub-region and providing endurance support.

1) *Objective Functions*: To provide a comprehensive evaluation of the HDDPP model, two objective functions are employed for measurement. These are distance cost, and the number of end drones amounted.

- *Distance Cost (Objective function 1)*: As cost is the primary objective of logistics companies aiming for energy savings and efficiency improvements, the distance cost is taken into consideration as the first objective function of the HDDPP, as denoted in (1).

$$\min : f_1 = w_1 \sum_{i \in C^l \cup D} \sum_{j \in C^l \cup D} d_{i,j} x_{i,j} + w_2 \sum_{k=1}^K \left(\sum_{i \in C_k} \sum_{j \in C_k \cup A} d_{i,j} y_{i,j,k} \right). \quad (1)$$

where w_1 represents the unit distance cost of mother drone, and w_2 represents the unit distance cost of small drone.

- *The Number of End Drones (Objective function 2)*: Determining the best number of end drones is a crucial factor in the HDDPP. Overutilization of end drones can result in elevated distribution costs, while an inadequate number of end drones may not suffice to complete all distribution tasks due to power and range constraints. Hence, the number of end drones is considered the second objective function of

TABLE I
SYMBOLS AND DEFINITIONS

Symbols	Definitions
<i>Sets:</i>	
G	$= (V, E)$. V is the node-set, E is the set of edges between two nodes;
V	$= D \cup A \cup C$. D is the depot, A is the automatic airport set,
	C is the customer node set;
R	$= \{R_m, R_e\}$. The set of routes of mother drone and end drone;
R_m	The routes of the mother drone;
R_e	The routes of the end drones;
C	$= \{c_1, c_2, \dots, c_n\}$, the set of customer nodes;
C_k	The set of customer nodes in the k -th cluster, where $C_k \subseteq C$;
C_k^l	The node of the mother drone launches an end drone in the k -th cluster, where $C_k^l \in C_k$;
C^l	$= \{C_1^l, C_2^l, \dots, C_K^l\}$, the node set of the mother drone launches end drones, where $C^l \subseteq C$;
C_d	The customer nodes for delivery, where $C_d \subseteq C$;
C_p	The customer nodes for pickup, where $C_p \subseteq C$;
D	$= \{0, K+1\}$, the depot;
A	The set of automatic airports;
$P(x)$	$= \{P_1(x), \dots, P_{r-1}(x)\}$, the mixture probability density distribution;
<i>Params:</i>	
d_i	Delivery weight at customer node i ;
p_i	Pickup weight at customer node i ;
$d_{i,j}$	The distance between the node i and the node j , $i, j \in V$;
w_e	The weight of the drone;
$q_{i,k}$	The parcel's weight of the i -th customer in k -th cluster;
$Q_{i,k}$	The parcel weight carried by the end drone when visiting the i -th customer in k -th cluster;
n_k	The number of customers in the k -th cluster;
Q_S	The max payload capacity of an end drone
M_i	The mapping from the i -th source model to the target task instance;
ρ_1, ρ_2	The transportation costs per unit of distance of mother drone and end drones, respectively;
mt_i	The arrival time for the mother drone to the customer node i ;
$et_{i,k}$	The arrival time for the k -th end drone to the customer node i ;
$z_{i,j,k}^p$	The pickup weight at arc (i, j) in k -th cluster;
$z_{i,j,k}^d$	The delivery weight at arc (i, j) in k -th cluster;
<i>Vars:</i>	
$x_{i,j}$	$x_{i,j} = 1$, if the mother drone goes from node i to node j . otherwise, $x_{i,j} = 0$, where $i, j \in C^l \cup D$;
$y_{i,j,k}$	$y_{i,j,k} = 1$, if the k -th end drone goes from node i to node j in the k -th cluster. otherwise, $y_{i,j,k} = 0$, where $i \in C_k$, $j \in C_k \cup A$;

HDDPP, as denoted in (2).

$$\min : f_2 = N_{drones}. \quad (2)$$

2) *Constraints:* To more effectively capture the intricacies of the distribution scenario involving heterogeneous multi-drones, four distinct categories of constraints are meticulously considered: a) Constraints on the mother drone. b) Constraints on the end drones. c) The formulation of an energy consumption model specifically tailored for end drones. d) Constraints addressing the simultaneous execution of pickup and delivery operations. These categorized constraints collectively contribute to a comprehensive and nuanced representation of the complex interactions within the context of heterogeneous multi-drone distribution systems.

- a) *Mother drone constraints:* Assuming that the mother drone has sufficient payload and sufficient range, the mother drone should meet the following constraints.

$$\sum_{i \in C^l \cup D} x_{i,j} = 1, \quad \forall j \in C^l \cup D, \quad (3)$$

$$\sum_{j \in C^l \cup D} x_{i,j} = 1, \quad \forall i \in C^l \cup D, \quad (4)$$

$$\sum_{j \in C^l \cup D} x_{0,j} = 1, \quad (5)$$

$$\sum_{i \in C^l} x_{i,K+1} = 1, \quad (6)$$

$$\sum_{i \in C^l \cup D} \sum_{j \in C^l \cup D} x_{i,j} \leq K - 1, \quad (7)$$

$$\sum_{i \in C^l} x_{i,h} - \sum_{j \in C^l \cup D} x_{h,j} = 0, \quad \forall h \in C^l, \quad i \neq j, \quad (8)$$

$$x_{i,j} \in \{0, 1\}, \quad \forall i, j \in C^l \cup D, \quad (9)$$

The imposed constraints delineate specific requirements within the formulated model. Constraints (3)–(4) serve to restrict the mother drone to a singular visitation to each cluster, ensuring an unambiguous traversal pattern. Constraint (5) is designed to guarantee that the mother drone initiates its route from the designated depot. Subsequently, Constraint (6) mandates the mother drone's eventual return to the depot, completing the closed routing. To prevent the formation of sub-loops in the mother drone route, Constraint (7) is imposed. The sequential visitation of clusters is enforced by Constraint (8), ensuring that the mother drone departs from the h -th cluster after visiting it. Lastly, Constraint (9) delineates the permissible range of decision variables for the mother drone, contributing to the overall model precision.

- b) *End drone constraints:* End drones should meet constraints such as payload constraints and no sub-loop constraints etc. The specific definition is as follows.

$$\sum_{i \in C_k} y_{i,j,k} = 1, \quad \forall j \in A \cup C_k \setminus C_k^l, k \in \{1, 2, \dots, K\}, \quad (10)$$

$$\sum_{j \in C_k \cup A} y_{i,j,k} = 1, \quad \forall i \in C_k, k \in \{1, 2, \dots, K\}, \quad (11)$$

$$\sum_{i \in C_k} y_{i,h,k} - \sum_{j \in C_k \cup A} y_{h,j,k} = 0, \quad \forall h \in C_k, \quad i \neq j, \quad (12)$$

$$\sum_{i \in C_k} \sum_{j \in C_k \cup A} y_{i,j,k} \leq n_k - 1, \quad \forall k \in \{1, 2, \dots, K\}, \quad (13)$$

$$y_{i,j,k} \in \{0, 1\}, \quad \forall i \in C_k, j \in C_k \cup A, k \in \{1, 2, \dots, K\}, \quad (14)$$

$$q_{i,k} \geq 0, \quad \forall i \in C_k, k \in \{1, 2, \dots, K\}, \quad (15)$$

Constraints (10)–(11) have been formulated to impose restrictions on the visitation of each customer node by an end drone, ensuring a singular visitation to each customer node. Additionally, Constraint (12) is designed to ensure that an

end drone departs from the h -th node in the corresponding cluster after visiting it. The formulation of Constraint (13) serves the purpose of prohibiting sub-loops within the routes of end drones. Furthermore, Constraint (14) delineates the acceptable range for end drone decision variables, contributing to the precision of the overall model. The necessity for non-zero weights for each parcel is asserted by Constraint (15), ensuring that the weight of each parcel exceeds zero. Moreover, the collective weight of parcels within each cluster must surpass zero, guaranteeing that end drones are authorized to distribute parcels solely when there are parcels present within the cluster. This intricate system of constraints serves to govern the behavior of end drones with a nuanced consideration of the specificities associated with parcel distribution.

- *c) Energy consumption model of end drones:* The energy consumption model of end drones is developed according to the work of [42], and shown below,

$$En(Q_S) = (w_e + Q_S)^{\frac{3}{2}} \sqrt{\frac{g^3}{2\rho\zeta r_n}}, \quad (16)$$

$$e_i = En(Q_S) \cdot t, \quad \forall i \in C_k, \quad (17)$$

$$\sum_{i \in C_k} e_i \leq E_{\max} \quad \forall k \in \{1, 2, \dots, K\}, \quad (18)$$

In the above Equations, w_e symbolizes the self-weight of the end drone, ζ signifies the rotor area, ρ represents the air-fluid density, r_n denotes the number of rotors, g stands for gravity, and t signifies the flight time. The energy consumption model, denoted as $En(Q_S)$, is explicated in constraint (16). Notably, this energy consumption model exhibits an approximately linear relationship with its payload Q_S , as encapsulated in constraint (17). This linear dependence on payload represents a critical facet of the energy consumption dynamics associated with the end drone's operation.

In accordance with the aforementioned energy consumption model for an end drone, the imperative energy constraint that necessitates fulfillment is expressed in constraint (18). This constraint encapsulates the requirement that the end drone must adhere to in terms of its energy utilization, providing a guideline for the optimization framework.

- *d) Simultaneous pickup and delivery constraints:* The constraints of simultaneous pickup and delivery should satisfy the payload of end drones during a flight.

$$\sum_{j \in C_p} z_{i,j,k}^p - \sum_{j \in C_p} z_{j,i,k}^p = p_i, \quad \forall i \in C_k, \quad (19)$$

$$\sum_{j \in C_d} z_{i,j,k}^d - \sum_{j \in C_d} z_{j,i,k}^d = d_i, \quad \forall i \in C_k, \quad (20)$$

$$\sum_{i \in C_k} \sum_{j \in C_k} z_{i,j,k}^d * y_{i,j,k} + \sum_{i \in C_k} \sum_{j \in C_k} z_{i,j,k}^p * y_{i,j,k} \leq Q_S, \quad \forall k \in \{1, 2, \dots, K\}, \quad (21)$$

$$Q_{j,k} \geq Q_{i,k} - z_{i,j,k}^d + z_{i,j,k}^p - M(1 - y_{i,j,k}),$$

$$\forall i, j \in C_k, k \in \{1, 2, \dots, K\}, \quad (22)$$

Concerning the pickup and delivery constraints, their fulfillment is mandated by constraints (19) and (20), which rigorously enforce the conservation of traffic during pickup and delivery operations performed by the end drone. Moreover, due to the inherent weight capacity limitations of an end drone, it is imperative that the combined weight of parcels designated for pickup and delivery remains below the payload capacity of the end drone. Consequently, constraint (21) explicitly specifies that the total weight of parcels carried by an end drone must not exceed its designated payload capacity.

Furthermore, constraint (22) intricately dictates that during the delivery operation from customer node i to customer node j , the weight of goods carried by the end drone at customer node j must be greater than or equal to the sum of the weight at customer node i and the pickup weight at customer node i , or less the delivery weight at customer node i . This formulation ensures precise adherence to weight-related considerations during the entire pickup and delivery process, accounting for the dynamic changes in the cargo carried by the end drone.

IV. PROPOSED METHOD

In this section, we introduce the details of our TOHO, and explain how we utilize the existing knowledge from related source tasks to enhance the convergence in solving target tasks.

A. Framework of TOHO

The proposed TOHO mainly employs two strategies to generate offspring: 1) the transfer optimization with topology alignment strategy and 2) the ensemble genetic optimization strategy with heuristic operators. In each iteration, either the transfer optimization with topology alignment strategy, or the ensemble genetic optimization with heuristic operators is executed. Specifically, transfer optimization with topology alignment strategy is executed every δt iteration, with the aim of accelerating the convergence by reusing the individual probabilistic models of elite candidate solutions $P(\mathbf{x}) = \{P_1(\mathbf{x}), \dots, P_{T-1}(\mathbf{x})\}$ drawn from previously solved optimization tasks. In terms of the ensemble genetic optimization with heuristic operators, several mutation and crossover operators are designed based on the characteristics of HDDPP to explore a better solution further. Moreover, we also design a voting-based ensemble strategy to select the Pareto frontier with high-quality diversity and convergence. To illustrate the proposed TOHO more clearly, we present its flowchart in Fig. 4. Furthermore, the pseudocode of TOHO is summarised in Algorithm 1.

B. Transfer Optimization Strategy With a Topology Alignment

In the proposed transfer optimization with topology alignment strategy, firstly, a transfer optimization strategy based on multiple probability distribution models is presented. Secondly, a Monte Carlo approximation strategy is designed to obtain transformed source models and update the probabilistic mixture model. Thirdly, a topological alignment strategy is proposed in

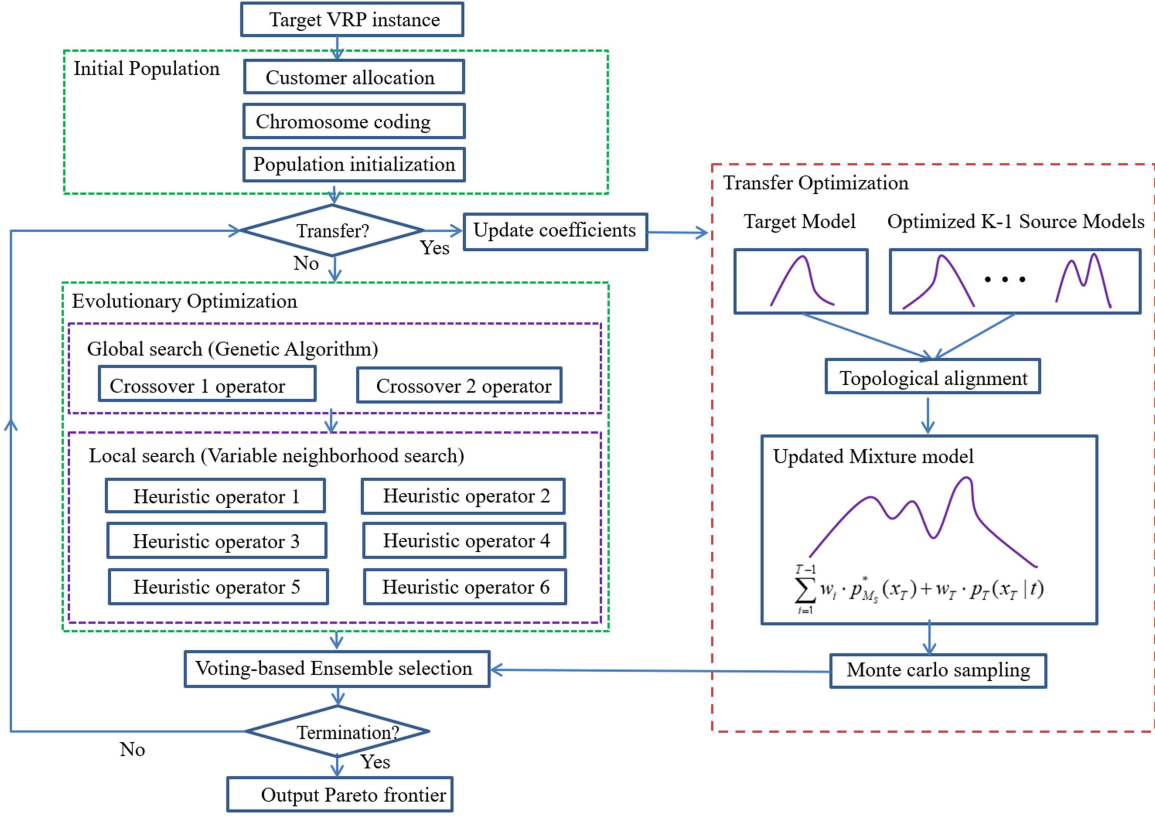


Fig. 4. The mixture model-based transfer optimizer with heuristic operators framework for HDDPP.

transfer optimization to induce the positive transfer of knowledge to a greater extent.

1) *Transfer Optimization Strategy*: In the TOHO, a transfer optimization strategy with multiple sub-models is developed. Let $\{T_1, T_2, \dots, T_{T-1}\}$ be the source tasks. For each source task, there is an individual probabilistic density distribution model of elite solutions, i.e., $P_i(x), i \in \{1, \dots, T-1\}$. We denote the target task, i.e., the instance that is to be optimized, as T_T , and its corresponding probabilistic density distribution model (formed by elite solutions) is represented as $P_T(x)$.

Based on the transfer optimization algorithm proposed by [14], a topological alignment strategy is proposed to induce the positive knowledge transfer from the source models to the target task instance. This strategy maps solutions in any source model from its original search space χ_i to the target task search space. It is essential to learn the spatial transformation of all the sub-models in the source mixture model to the target task instance to achieve topological alignment.

Thus, the probabilistic model-based transfer optimization with topology alignment strategy can be expressed as (23).

$$\begin{aligned}
 & \max_{w_1, w_2, \dots, w_T} \int_{\chi_T} F_T(x_T) \left[\sum_{i=1}^{T-1} w_i \cdot P_{M_i}(x_i) + w_T \cdot P_T(x_T) \right] dx_T \\
 & s.t., \quad \sum_{i=1}^{T-1} w_i + w_T = 1, \\
 & \forall w_i, w_T \geq 0, \quad i \in \{1, 2, \dots, T-1\},
 \end{aligned} \tag{23}$$

where $x_T \in \chi_T$ is the candidate solution of the target task instance T . $P(x)$ is the probabilistic density distribution of the solution x . $M_i: \chi_i \rightarrow \chi_T$ is the mapping from the i -th source model to the target task instance, $P_{M_i}(x_T)$ is the transformation of $P_i(x_i)$ under mapping M_i . $F_T(x) = (f_1(x), f_2(x), \dots, f_m(x))$ is a multi-objective function with m objectives. w_i is transfer coefficient of source sub-models, where $i \in \{1, \dots, T-1\}$, and satisfies $\sum_{i=1}^T w_i = 1$, and $w_i > 0$.

From (23), the transfer coefficient w_i represents the transfer degree of the i -th source sub-model. A higher value of w_i indicates a stronger overlap between the search distribution of the i -th source sub-model $P_i(x_i)$ and the target task instance, which is classified as a positive transfer of knowledge from the i -th source sub-model to the target task instance. Conversely, a smaller value of w_i indicates significant differences between the search distribution of the i -th source sub-model and the target task instance, thus reducing the efficacy of knowledge transfer from the i -th source sub-model. Essentially, the key concept behind the TOHO is that if there exists a source model T_i such that $P_i(x) \neq P_T(x)$, where $P_T(x)$ is the distribution of the target task, setting w_i close to 1 and sampling from the source model will immediately generate a high-quality solution for the target task T_T .

The probabilistic-model-based Transfer Optimization (TOHO) has a relatively small memory footprint and low time complexity. Considering a scenario involving a source task where a significant number (denoted as N) of solutions has been evolved, each comprising b binary bits. The straightforward

Algorithm 1: The Transfer Optimization With Heuristic Operators (TOHO).

Input: Individual probabilistic density distribution models of elite solutions drawn from source tasks $P(\mathbf{x}) = \{P_1(\mathbf{x}), \dots, P_{T-1}(\mathbf{x})\}$; customer node-set C ; mutation operator set N_o ;

Output: The optimized population p'_a ;

- 1 Initialize the target population p_a according to the designed chromosome coding mode in Fig. 5;
 - 2 The objective functions f_1 and f_2 in the model are used to evaluate the population;
 - 3 The task assignment strategy based on clustering is used to assign customer nodes to the end drone;
 - 4 The clustering adjustment operators designed in Fig.7 is used to adjust the infeasible task allocation scheme;
 - 5 **while** *termination criterion is not met* **do**
 - 6 **if** $\text{mod}(t, \delta t) = 0$ **then**
 - 7 # Transfer optimization strategy ;
 - 8 $o_T \leftarrow$ Call transfer optimization strategy (Algorithm 2) to generate offspring population;
 - 9 **else**
 - 10 # Heuristic strategy ;
 - 11 $o_T \leftarrow$ Use the crossover operator from N_o designed in Fig.8 is used to cross the solutions;
 - 12 $o_T \leftarrow$ Use the operators from N_o designed in Fig.9 is used to mutate the solutions;
 - 13 Evaluate $2N$ solutions of parent and offspring populations;
 - 14 $p'_a \leftarrow$ Call voting-mechanism based ensemble selection strategy (Algorithm 4) to select the best N solutions from the parent and offspring populations as the next generation population;
 - 15 $t \leftarrow t + 1$;
-

storage of raw solution data would necessitate Nb bits of memory. In contrast, a factored Bernoulli distribution can encode higher-order knowledge about the underlying distribution of the same solution population, its complexity only $\mathcal{O}(\log_2 N)$ [43]. Regarding the assessment of the computational feasibility of the proposed TOHO, it is observed that the algorithm typically demonstrates rapid convergence, often within the initial iterations. In other words, incorporating the entire learning algorithm as a nested module within any external Evolutionary Algorithm is reasonable, incurring minimal computational overhead.

Furthermore, according to [14], suppose χ represents a unified search space where all source models and target tasks are defined. Therefore, the target mixture model can be constructed by optimal superposition of the source task model and target task model in unified space χ . In particular, the probabilistic mixture model-based transfer coefficient w_i can be obtained by solving the following log-likelihood maximizer in (24):

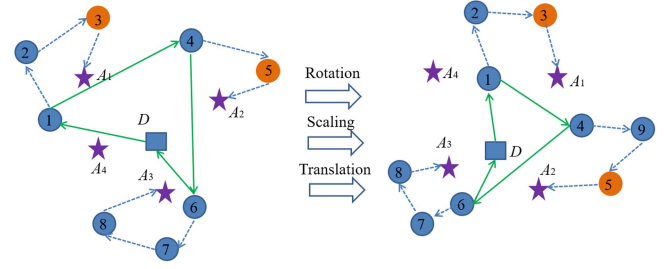


Fig. 5. Topology alignment strategy for solutions.

Algorithm 2: Transfer Optimization Strategy.

Input: The source mixture models $P(\mathbf{x})$; topological alignment mapping M_i

Output: Transferred offspring population o_T ;

- 1: $M_i \leftarrow$ Use parent population p_a and prior knowledge from the mixture model to learn the topological alignment mapping $\chi_i \rightarrow \chi_T$ between instances according to (25) and (26);
 - 2: $P_{M_i}(\mathbf{x}_T) \leftarrow$ Call the Monte Carlo approximation strategy (Algorithm 3) to update the transformed source model with M_i ;
 - 3: $w = \{w_1, w_2, \dots, w_T\} \leftarrow$ Update the weight vector of the mixed model and build the mixture model-based transfer optimization model according to $P_{M_i}(\mathbf{x}_T)$ and $P_T(\mathbf{x}_T|t)$;
 - 4: $o_T \leftarrow$ Sample from the mixture model to generate the offspring population;
-

$$\max_{w_1, w_2, \dots, w_T} \sum \log \left(\sum_{i=1}^{T-1} w_i \cdot P_{M_i}(\mathbf{x}_i) + w_T \cdot P_T(\mathbf{x}_T|t) \right), \quad (24)$$

Upon learning the appropriate mapping M_i , it is observed that during knowledge transfer ($w_i > 0$), sampling from the probabilistic mixture model $\text{mix}M(\mathbf{x})$ obtained from (24) can be an even approximate optimal solution. This means that TOHO activates the positive transfer of knowledge by utilizing mapping M_i from the source model to the target task instance, while simultaneously reducing the transfer of harmful knowledge by sampling the probabilistic mixture model. Thus, the procedure of the transfer optimization strategy is summarized in Algorithm 2.

2) *Transformed Source Model by Monte Carlo Approximation:* From Algorithm 2, the transformed source models $P_{M_i}(\mathbf{x}_T)$ can be obtained by Monte Carlo approximation. The procedure of the Probabilistic mixture model-based Monte Carlo approximation strategy is summarised in Algorithm 3.

From Algorithm 3, the Monte Carlo approximation strategy adopts the probability distribution model of the mixed model and Mapping M_i to obtain the transformed source models $P_{M_i}(\mathbf{x}_T)$, and these sources models make up the probabilistic mixture model $\text{mix}M(\mathbf{x})$ according to $\text{mix}M(\mathbf{x}) =$

Algorithm 3: Probabilistic Mixture Model-Based Monte Carlo Approximation Strategy.

Input: The source mixture models $P(x)$; topological alignment mapping M_i

Output: Transformed source models $P_{M_i}(x_T)$;

```

1 for each source model  $P_i(x_i)$  do
2    $x_i \leftarrow$  Sample  $N$  solutions from the  $i$  source model
   search space  $\chi_i$ ;
3    $x_T \leftarrow M_i(x_i)$  Obtain  $N$  transformed solutions
   according to the search space mapping  $\chi_i \rightarrow \chi_T$ ;
4    $P_{M_i}(x_T) \leftarrow$  Update the probabilistic density
   distribution model to build the transformed source
   model;

```

$\sum_{i=1}^{T-1} w_i \cdot P_{M_i}(x_i) + w_T \cdot P_T(x_T|t)$. In this way, the probabilistic mixture model $mixM(x)$ is more likely to generate high-quality solutions for the target task instance T_T .

3) *Topology Alignment Strategy for Solutions:* When addressing diverse instances of the HDDPP in practical applications, the search spaces of the source model and the target task instances are denoted as χ_i (with dimension d_i) and χ_T (with dimension d_T), respectively. Due to the fact that the source and target search spaces may have different dimensions (i.e., $d_i \neq d_T$), we propose the topology alignment strategy to transform the individual probabilistic models into a unified search space. Moreover, the proposed topology alignment strategy also has the capability to increase the overlap between the distributions of elite solutions the source and the target instances, thereby inducing positive transfer while reducing negative transfer from irrelevant prior knowledge.

The proposed topological alignment strategy contains the following two steps:

Step 1: We learn the mapping M_i that aligns the geometric distribution of customer nodes in the source model with that of the target instance to the greatest extent possible. In the topology alignment strategy, the routing diagram of the source model is first scaled to match the geometric dimensions of the routing diagram of the target task instance. Subsequently, translation and rotation operations are performed to align the centroid and principal components of the customer distribution in the source model with those of the target task instance, as illustrated in Fig. 5. The spatial transformation from the source model to the target task instance can be mathematically defined using (25) and (26), through which the mapping matrix M_i can be learned.

$$M_i(u_{i,j}) = \alpha B \cdot u_{i,j} + b, \forall j \in \{1, 2, \dots, d_i\}$$

$$s.t., \quad \alpha = \frac{\sum_{k=1}^{d_T} t_{T,k}/d_T}{\sum_{j=1}^{d_i} t_{i,j}/d_i}, \quad (25)$$

$$B = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}, \quad (26)$$

where vector $u_{i,j}$ represents the Cartesian coordinates of node $j \in \{1, \dots, d_i\}$, while $\alpha \in R^C$ denotes the scaling constant and B denotes the angular rotation matrix. The angle θ denotes

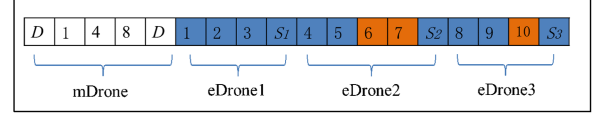


Fig. 6. Chromosome coding mode (This chromosome contains three end drone routes and one mother drone route, with blue for the delivery nodes and orange for the pickup nodes).

the angular difference between the principal components of the source and target instances, and $b \in R$ denotes the translation vector.

Step 2: We duplicate and re-assign the node index numbers in the source model. In the case where the number of nodes in the source model exceeds that of the target task instance ($d_i > d_T$), the surplus nodes ($d_i - d_T$) are randomly assigned, without replacement, to nodes within the set $j \in \{1, \dots, d_i\}$.

By employing the aforementioned topology alignment strategies, a closer alignment between the customer node distributions in the routes of the source model and the target instance is achieved. Consequently, the learned topological relationship matrix M_i will more likely capture the shared knowledge regarding optimized routes or sub-routes between the source model and the target instance. Fig. 5 provides a schematic representation of the devised topological alignment strategy. Notably, considering the potential lack of geometric correspondence between the source model and the routes of the target instance in Fig. 5, the mapping relationship matrix M_i aims to match the geometric distribution of customer nodes. Furthermore, it is important to highlight that the mapping relationship matrix M_i preserves the shortest path, thereby ensuring that the TOHO does not alter the optimal route in the transformed space [44].

C. Ensemble Genetic Optimization Strategy With Heuristic Operators

In the iterations without triggering transfer optimization, an ensemble genetic optimization strategy with heuristic operators is designed. This strategy consists of four main modules: 1) Designing the chromosome coding mode, where visiting customer nodes of the mother drone and several end drones are subject to integer coding; 2) Developing a customer nodes task assignment strategy for end drones considering the payload of end drones; and 3) Designing six heuristic operators based on the characteristics of HDDPP and the properties of pick-up nodes and delivery nodes; 4) Developing a voting-based ensemble selection strategy to select high-quality Pareto frontier.

1) *Chromosome Coding Mode Design:* The encoding mode of chromosomes could significantly influence the genetic-based algorithm's performance, hence, it is crucial to develop an effective chromosome encoding mode. In the encoding and decoding of HDDPP, the order of the launching nodes of the mother drone, the delivery nodes of the end drones, the pickup nodes of the end drone, and the automatic airport recycle nodes should be considered. To accomplish this, an integer coding approach is designed in which each gene represents a label of a customer node to encode chromosomes. Fig. 6 illustrates the chromosome

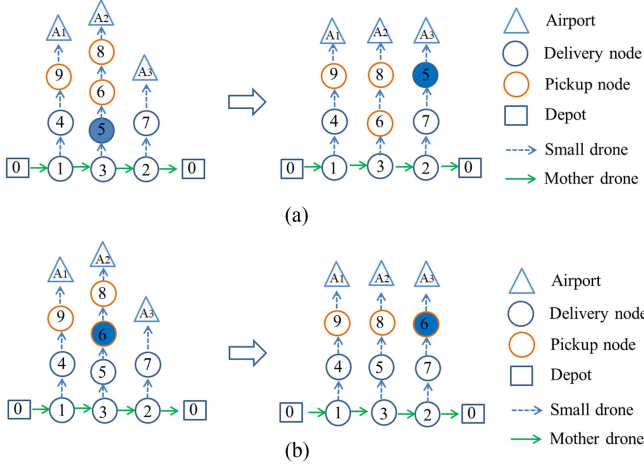


Fig. 7. Heuristic operators for adjustment of task assignment. (a) Deletion-insertion operators for delivery customer nodes. (b) deletion-insertion operators for pick-up customer nodes.

coding diagram for an instance that includes a route of a mother drone and three end drone routes comprising delivery nodes (blue) and pick-up nodes (orange).

2) *Customer Nodes Assignment Strategy*: In the HDDPP, the payload constraint of the end drone is determined by giving priority to the total weight of the customer nodes that need to be delivered, and then considering that the total weight of the customer nodes to be picked up cannot exceed the payload of the end drone. To refine the task allocation scheme, the total weight of customer nodes requiring delivery is initially considered during clustering. However, the payload of the end drone may be exceeded due to varying pickup and delivery sequences during routing. To address this, several clustering adjustment operators are designed to modify infeasible solutions that violate the payload constraint.

Specifically, an initial customer assignment for end drones is generated prior to routing optimization. To achieve this, the customer distribution region is divided into K sub-region and assigned to each end drone such that each sub-region corresponds to an end drone. Considering the payload constraints of the end drone, an improved task assignment strategy based on clustering is proposed.

Given the payload constraint of the end drones, the initial task assignment scheme generated by the task assignment strategy may result in the total weight of pick-up and delivery parcels exceeding the end drone's payload during the routing. To address this issue, two adjustment heuristic operators for task allocation are designed based on the payload constraint of the end drone. These operators include deletion-insertion operators for the delivery customer node or the pick-up customer node, respectively, with the following functionalities: 1) deleting a delivery node and inserting it into another cluster, see in Fig. 7(a), and 2) deleting a pickup node and inserting it into another cluster, see in Fig. 7(b).

3) *Specific Heuristic Operators Design*: In the designed genetic optimization strategy, several specific heuristic operators are designed According to the characteristics of HDDPP to

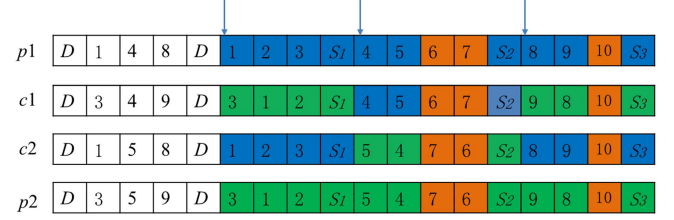


Fig. 8. Schematic diagram of multi-point crossover operator.

optimize the routes of the mother drone and end drones. The designed heuristic operators, including specific crossover operators and specific mutation operators, are depicted in Fig. 7 and Fig. 8, respectively.

a) *Specific crossover operator design*: A multi-point crossover operator is designed for crossing sub-region, wherein each sub-region corresponds to a crossover point, resulting in a K -point crossover operator. The coding in Fig. 6 is taken as an example, and the K -point crossover diagram of the sub-region is depicted in Fig. 8. The parent solution has three intersections, namely p_1 (1, 4, 8) and p_2 (3, 5, 9), which leads to two offspring solutions, namely o_1 and o_2 . If the resulting solution is infeasible, the parent solution is retained. Using this method, N solutions can be generated, where the number of intersections depends on the number of sub-region.

b) *Specific mutation operators design*: Six specific mutation operators have been designed to optimize the routes of the mother drone and the end drones. These include customer node exchange, either within the same cluster or between different clusters, as well as pick-up node and delivery node exchange within the same cluster. The details of these mutation operators are presented below:

Mutation operator 1: Exchange two delivery nodes of an end drone in different clusters, as illustrated in Fig. 9(a).

Mutation operator 2: Exchange a delivery node and a pickup node of end drones in different clusters, as illustrated in Fig. 9(b).

Mutation operator 3: Exchange two delivery nodes of an end drone in a cluster, as illustrated in Fig. 9(c).

Mutation operator 4: Exchange the pick-up nodes of an end drone in a cluster, as illustrated in Fig. 9(d).

Mutation operator 5: Exchange two mother drone nodes in different clusters (the mother drone nodes are the nodes launched end drones), as illustrated in Fig. 9(e).

Mutation operator 6: Exchange a mother drone node and an end drone node in a cluster, as illustrated in Fig. 9(f).

4) *Voting-Based Ensemble Selection Strategy*: Since multiple objective functions are given in the HDDPP model, to robustly select the Pareto frontier with promising convergence and diversity, a voting-mechanism-based ensemble selection strategy is proposed [45], [46]. The strategy integrates multiple sorting methods, with each sorting method serving as a voter that votes among the $2N$ solutions in the combination of offspring and parent population. The most suitable N solutions with high-quality of diversity and convergence are selected as the Pareto frontier, which becomes the parent population of the next generation, and continues the iterative evolution until the termination condition is met. To illustrate the process of the

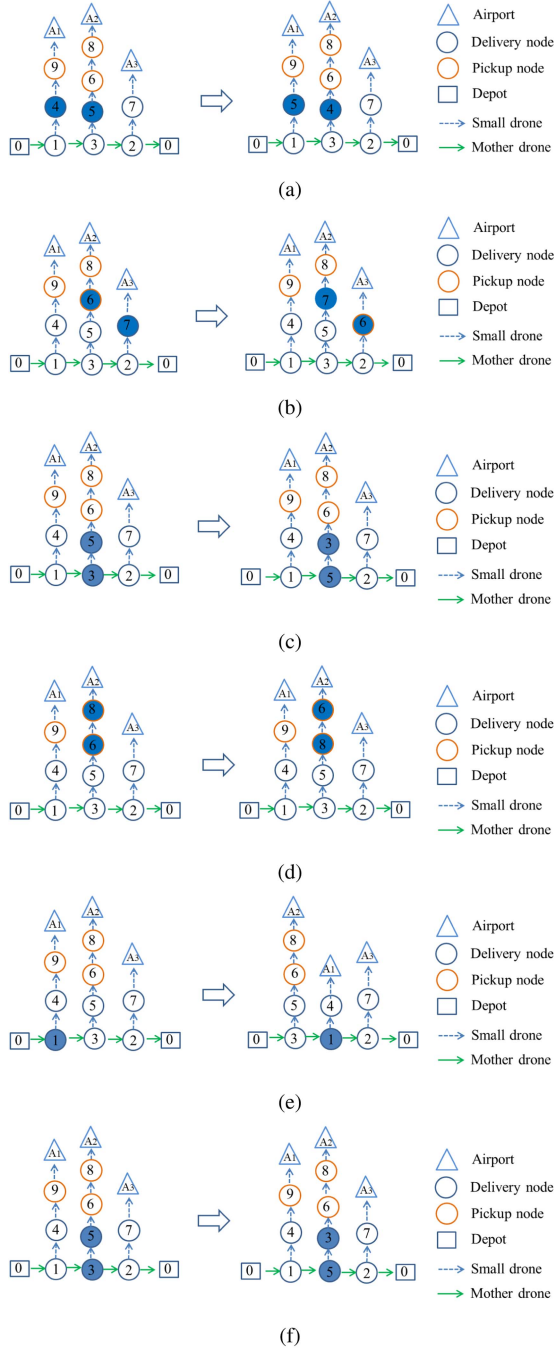


Fig. 9. Specific mutation operators. (a) Mutation operator 1. (b) Mutation operator 2. (c) Mutation operator 3. (d) Mutation operator 4. (e) Mutation operator 5. (f) Mutation operator 6.

voting mechanism, a schematic diagram is presented in Fig. 10. The main pseudocode of the voting-based ensemble selection strategy is summarised in Algorithm 4.

In Algorithm 4, the voting-based ensemble strategy comprises four main steps. Firstly, the parent and offspring solutions in the combined population are voted based on multiple sorting components in the S_{sort} (lines 1–9). Secondly, the weighted votes of each individual in the combined population are calculated (line 11). Thirdly, the most suitable N solutions in the combined population are selected based on their weighted votes $V_T(i)$ and

Algorithm 4: Voting-Based Ensemble Selection Strategy.

Input: Parent population p_a ; offspring population off ; the set of sorting method set S_{sort} ;
Output: The updated population p'_a ;

- 1 # Voting-based ensemble selection strategy;
- 2 Merge the population $p_a \cup off$;
- 3 **for** $i = 1 \rightarrow |S_{sort}|$ **do**
- 4 **for** $j = 1 \rightarrow |p_a \cup off|$ **do**
- 5 # Use the i -th sorting method $S_{sort}(i)$ to vote solutions in $p_a \cup off$;
- 6 **if** $S_{sort}(i)$ votes for the solution $p_a \cup off(j)$ **then**
- 7 $V(i, j) \leftarrow 1$;
- 8 **else**
- 9 $V(i, j) \leftarrow 0$;
- 10 # Calculate weighted votes;
- 11 $V_T(i) \leftarrow \sum_{j=1}^{|S_{sort}|} w_i \cdot V(i, j)$ Calculate the weighted votes of the i -th solution in all voters;
- 12 $F_p \leftarrow$ Calculate the crowding distance;
- 13 $p'_a \leftarrow$ Select the top N solutions according to the weighted votes $V_T(i)$ and crowding distance F_p ;
- 14 # Update voter weights;
- 15 **for** $i = 1 \rightarrow |S_{sort}|$ **do**
- 16 $S_{Vi} \leftarrow$ Calculate the number of solutions that are voted by the i -th sorting method and finally survive to the population p'_a ;
- 17 **for** $i = 1 \rightarrow |S_{sort}|$ **do**
- 18 $Con_i \leftarrow \frac{S_{Vi}}{\sum_{i=1}^{|S_{sort}|} S_{Vi}}$ Calculate contributions of the i -th voter based on S_{Vi} ;
- 19 # Adjust the voter weights based on their contributions;
- 20 **for** $i = 1 \rightarrow |S_{sort}|$ **do**
- 21 $w_i \leftarrow Con_i$;

crowding distance F_p (lines 12–13). Fourthly, the voters' weight is updated based on their contributions (lines 14–21).

To begin with, each sorting method is extracted from a multi-objective optimization algorithm (line 4). The votes obtained from the sorting method (the voter) are considered to have been selected by the corresponding environment selection method. Additionally, each sorting method has the same initialization weight. The weighted votes of the i -th solution $V_T(i)$ are calculated based on the voting results and their weights (line 11). The solution set is then divided into several layers based on the weighted votes, and the most suitable N solutions are selected from the population P based on the divided levels and their crowding distance (lines 12–13). Then, each voter's contribution Con_i is calculated based on the voting result V and the surviving individual S_{Vi} (lines 17–18), and the voter weights are updated based on their contribution during the optimization process (lines 19–21).

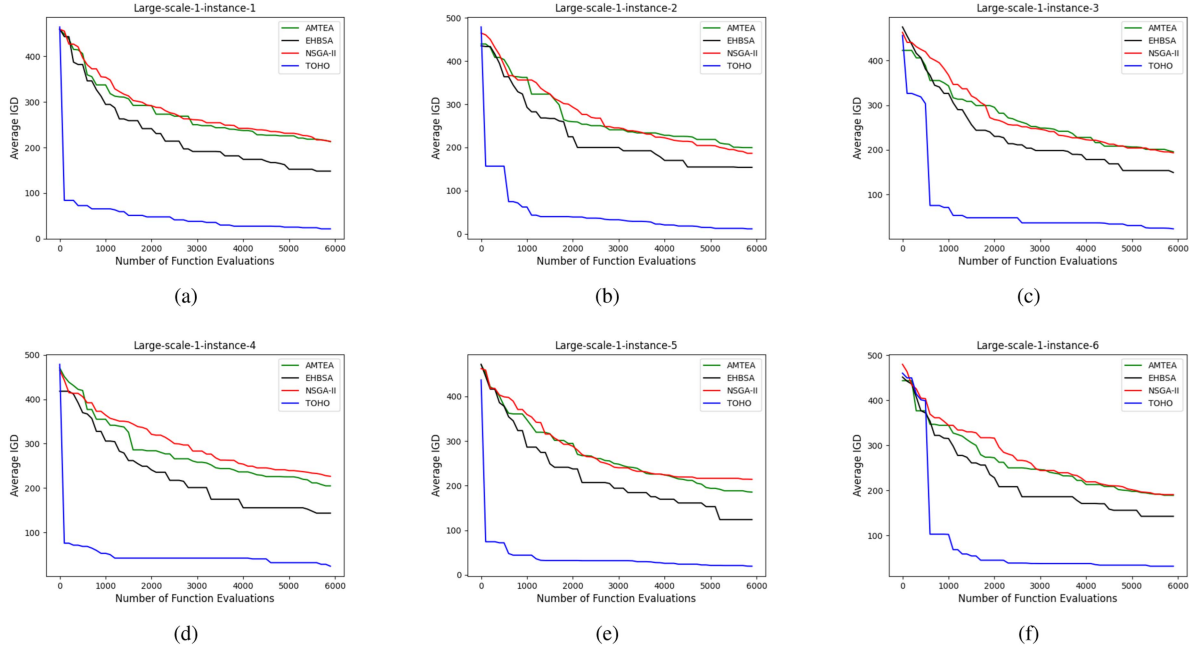


Fig. 10. The experimental results on instances of large-scale-1 (130 nodes). (a) Large-scale-1-instance-1. (b) Large-scale-1-instance-2. (c) Large-scale-1-instance-3. (d) Large-scale-1-instance-4. (e) Large-scale-1-instance-5. (f) Large-scale-1-instance-6.

V. EXPERIMENTS AND ANALYSIS

To investigate the performance of the proposed TOHO, it is compared with three other advanced baselines on a wide range of extended benchmark instances. All baseline implementations are scripted in Python and executed on a computing environment utilizing a 64-bit Windows 10 operating system, where the computational infrastructure is hosted on an Intel Core (TM) i7-9800X server, operating at a clock speed of 3.80 GHz.

A. Experiment Settings

1) *Target Instance Settings*: In this experiment, the HDDPP instances of source problem and target problem are extended from three public datasets, i.e., *vrpnc4*-150 with 130 customer nodes [47], *X-n214* with 200 customer nodes [48], and *Golden-8* with 270 customer nodes [49], which are extended in the following two ways: 1) Arbitrarily transforming the geometric distribution of customer nodes and randomly assigning index numbers to each customer node; 2) Removing customer nodes that exceed the maximum range according to the constraint of the end drones. These approaches allowed for the generation of diverse instances for evaluation. The extended instances are marked large-scale-1, large-scale-2, and large-scale-3, respectively.

2) *Drone Parameter Settings*: The parameter values of the mother and end drones are based on the specifications of real commercial drones. The specific drone parameter settings are given in Table II.

3) *Baseline Settings*: To evaluate the effectiveness of the TOHO based on topological alignment strategy, we compared

TABLE II
DRONE PARAMETER SETTINGS

$Q_s = 10kg$	The payload capacity of the end drone.
$r_n = 6$	The number of rotors.
$\mu = 1.204kg/m^3$	The fluid density.
$\zeta = 0.2m^2$	The rotor disc area.

it with three baselines: a well-known multi-objective optimization algorithm, i.e., a) NSGA-II optimization algorithm; and two transfer optimizers based on probabilistic mixture models, namely, b) Edge histogram sampling algorithm for multi-objective optimization (EHBSA) [50], which is a probabilistic model-based optimization algorithm for combination optimization problems; c) Transfer optimization algorithm (AMTEA) without topological alignment strategy [37]. We set the population size N of all baselines to 100, the crossover probability of genetic operator parameters to 0.75, and the mutation probability of exchange sequence to 0.2 [51]. For the EHBSA, we used the template-free sampling strategy and its default parameter settings. Additionally, we used the same heuristic operator settings in the baselines as those used in the proposed TOHO method.

4) *Metric Settings*: In this experiment, the Inverted Generational Distance (IGD) metric is employed to assess the proposed TOHO and the baselines. The IGD metric enables evaluating both the convergence and diversity of the population obtained by the baselines.

B. Benchmark Instance Test

The study conducted an experimental test on three large-scale instances of HDDPP, namely large-scale-1, large-scale-2, and

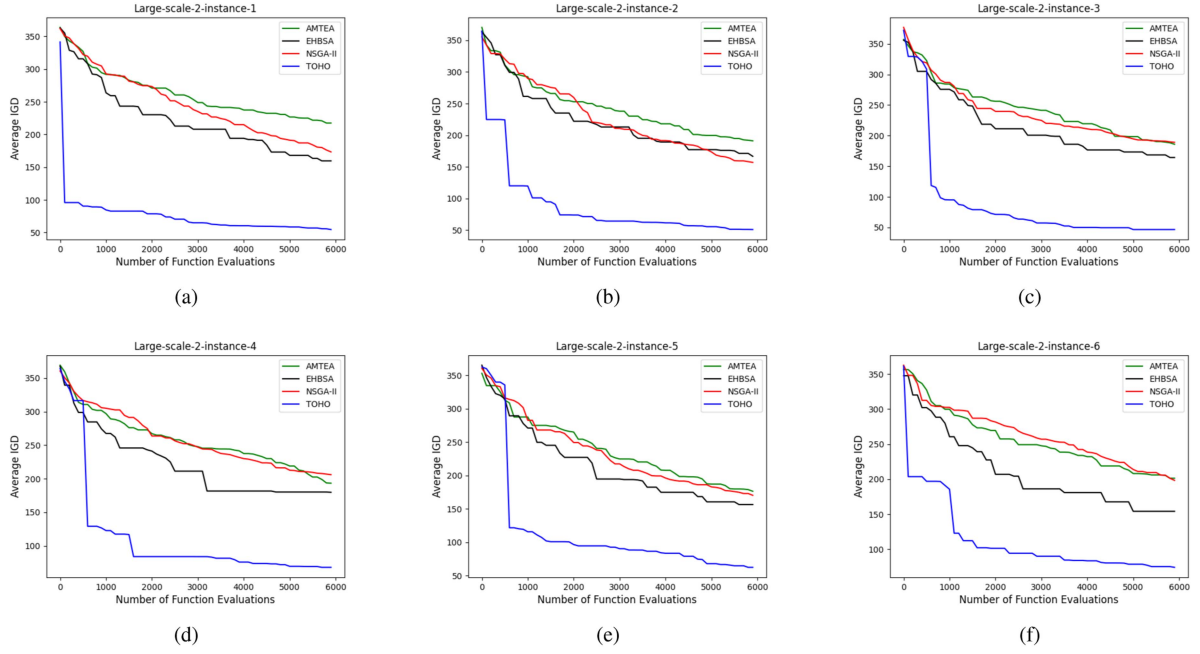


Fig. 11. The experimental results on instances of large-scale-2 (200 nodes). (a) Large-scale-2-instance-1. (b) Large-scale-2-instance-2. (c) Large-scale-2-instance-3. (d) Large-scale-2-instance-4. (e) Large-scale-2-instance-5. (f) Large-scale-2-instance-6.

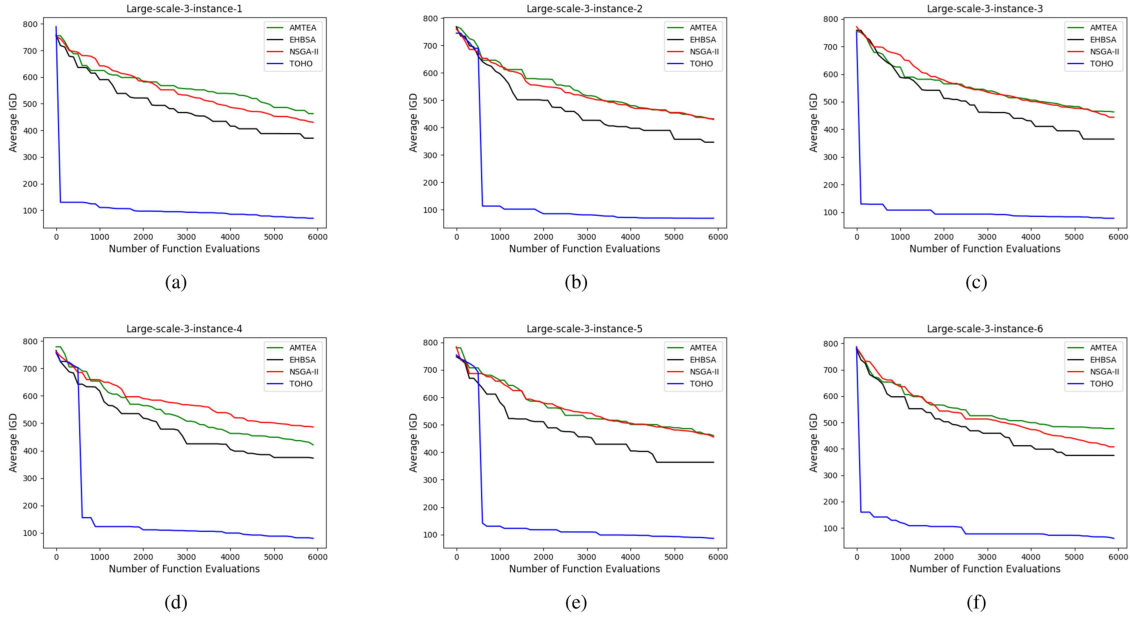


Fig. 12. The experimental results on instances of large-scale-3 (270 nodes). (a) Large-scale-3-instance-1. (b) Large-scale-3-instance-2. (c) Large-scale-3-instance-3. (d) Large-scale-3-instance-4. (e) Large-scale-3-instance-5. (f) Large-scale-3-instance-6.

large-scale-3. The objective of the experiment is to evaluate the optimization and convergence performance of four baselines, namely AMTEA, EHBSA, NAGS-II, and TOHO, in both source and target task instances with similar distributions. The convergence curves of these baselines are plotted in Figs. 10 to 12, where the y -axis represents the multi-objective IGD values obtained and the x -axis represents the number of evaluations conducted for each instance.

The performance evaluation depicted in Figs. 10 to 12 underscores the superior efficacy of the proposed TOHO compared to other baselines across HDDPP instances of varying scales: large-scale-1 (130 customer nodes), large-scale-2 (200 customer nodes), and large-scale-3 (270 customer nodes). Notably, TOHO exhibits a significantly accelerated convergence rate during the initial optimization stages. This notable achievement can be attributed to the activation of positive transfer mechanisms

within TOHO, facilitating the acquisition of high-quality solutions through effective alignment between the source and target instances. Specifically, following sampling from the mixture model, optimized knowledge from the source instance is seamlessly infused into the target instance, optimizing it through topological route alignment. Comparative analysis against other transfer optimizer baselines reaffirms TOHO's fast convergence rate.

Moreover, the AMTEA demonstrates the second-best performance across the three large-scale instances. However, it underperforms in four specific instances (large-scale-2-instance-1, large-scale-3-instance-1, large-scale-3-instance-5, and large-scale-1-instance-1). The underlying reasons for this suboptimal performance can be attributed to several factors: firstly, the absence of a solution topological alignment strategy renders the baseline susceptible to negative transfers arising from inconsistent topological relationships between the source and target tasks. Secondly, the adoption of a selection strategy based on a voting mechanism in TOHO enhances the identification of Pareto frontier solutions characterized by superior diversity and convergence properties. Lastly, the heuristic operator devised based on problem-specific characteristics in TOHO endows the algorithm with efficient search capabilities, contributing to its overall performance.

Furthermore, the analysis spanning from Figs. 10 to 12 underscores the remarkable effectiveness of the TOHO in harnessing the knowledge gleaned from optimized related tasks to enhance the routing performance of HDDPP instances. These findings attest to TOHO's adeptness in leveraging similar routing schemes from source tasks by extracting valuable knowledge embedded within the original search space of those tasks. Consequently, this experimental investigation underscores the potential of the topological alignment strategy to induce positive transfers of knowledge between disparate yet related instance tasks.

However, despite the notable success of TOHO in leveraging knowledge from similar tasks, its utilization of a generic search mechanism across different tasks imposes constraints on the rate of convergence improvement, even on highly similar HDDPP instances. To mitigate this limitation, TOHO employs a transfer optimizer to seamlessly transfer high-quality solutions from the optimized source task to the target task. Additionally, as the degree of similarity diminishes, TOHO's convergence rate surpasses that of the AMTEA on the target task of HDDPP.

Additionally, from Figs. 10 to 12, it is noteworthy that TOHO consistently outperforms NSGAI and EHBSA in terms of convergence rate across all instances. This superior performance can be attributed to two primary factors: firstly, the Monte Carlo sampling approximation algorithm within the mixed probability model adeptly procures solutions exhibiting high degrees of similarity. Secondly, transfer optimization strategies play a pivotal role in fostering the forward transfer of knowledge, thereby facilitating the acquisition of high-quality initial or intermediate solutions. These observations underscore the efficacy of TOHO in achieving superior convergence rates and highlight its potential for enhancing optimization performance across diverse problem instances.

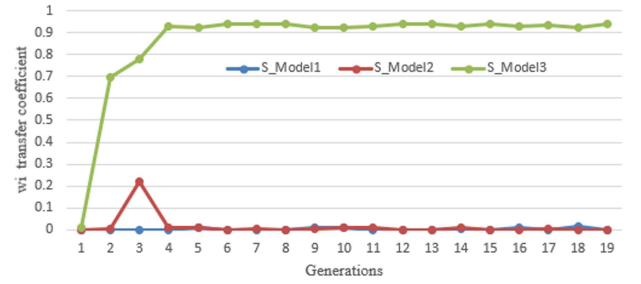


Fig. 13. The transfer coefficient of TOHO in a mixture model in large-scale-1.

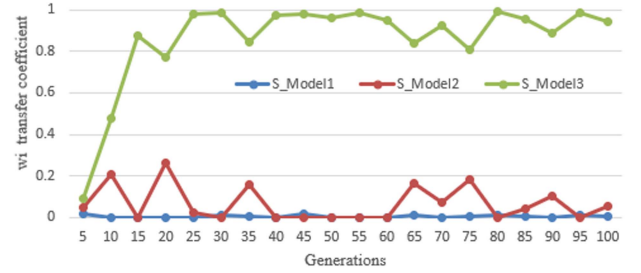


Fig. 14. The transfer coefficient of TOHO in a mixture model in large-scale-2.

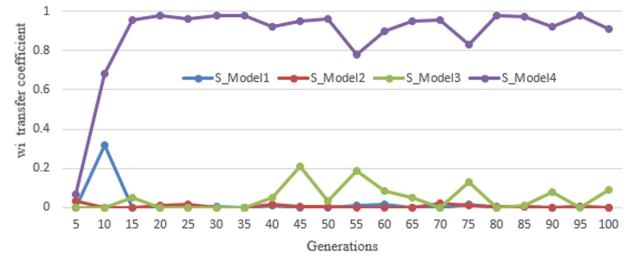


Fig. 15. The transfer coefficient of TOHO in a mixture model in large-scale-3.

To conduct a more in-depth analysis of the knowledge transfer between source models and target tasks in the mixture model, we conducted a statistical analysis of the curves of transfer coefficient w_i during the evolutionary process. A larger transfer coefficient indicates a greater degree of positive transfer from the source model to the target task, while a smaller transfer coefficient indicates a lesser degree of knowledge transfer. We present the curves of the knowledge transfer coefficient w_i in the mixture model for the three-scale instances of HDDPP in Figs. 13 to 15. The abscissa represents the number of iterations, while the ordinate represents the value of the transfer coefficient w_i . Each curve corresponds to the degree of knowledge transfer for different source models.

As depicted in Figs. 13 to 15, the TOHO exhibits significant variations across different source models. Specifically, in the large-scale-1 instances shown in Fig. 13, the mixture model incorporates three distinct source models (source sub-model 1, source sub-model 2, and source sub-model 3). The transfer coefficient w_i of source sub-models 1 and 2 are found to be relatively low, approaching zero, suggesting that the target task acquires limited knowledge and triggers minimal positive transfer. Conversely, source model 3 yields a larger transfer

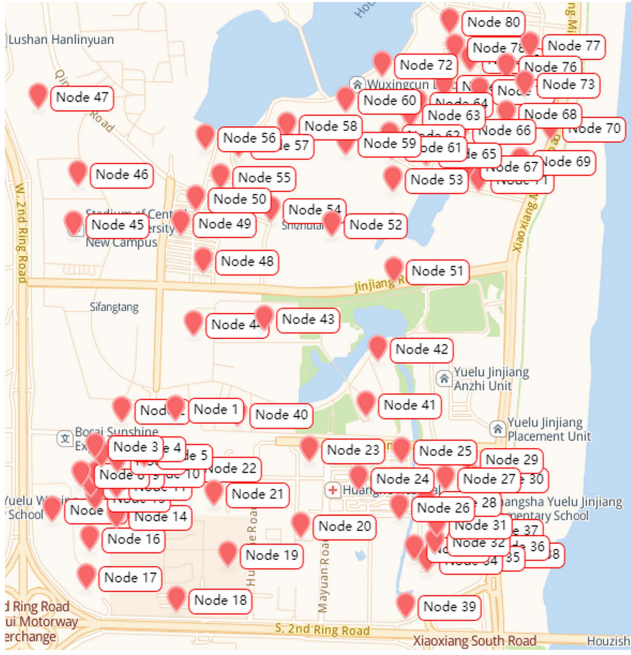


Fig. 16. The real-world instance includes 80 customer nodes in Changsha, China.

coefficient, indicating that the TOHO achieves topological relationship alignment between source sub-model 3 and the target task, resulting in significant topological overlap between source sub-models and the target task instance.

Similarly, the transfer coefficient curves of the large-scale-2 instances are shown in Fig. 14, also demonstrate a larger w_i for source sub-model 3, which corresponds to better alignment of the topological relationship between the target task and source sub-model 3, thereby enabling greater positive transfer. In contrast, the transfer coefficients for the other two source sub-model are smaller, indicating negligible positive transfer. The transfer coefficient curves of the large-scale-3 instances are displayed in Fig. 15, the mixture model includes four source models. Among these, the transfer coefficient learned in source sub-model 4 is the highest, yielding the maximum positive transfer of knowledge, while the transfer coefficient in source sub-model 3 is lower, leading to the acquisition of only partial knowledge. The transfer coefficients of source sub-models 1 and 2 tend toward zero, signifying that little valuable knowledge is gained from these source sub-model.

In summary, the topological alignment strategy employed by TOHO induces positive transfer and mitigates harmful negative transfer, enabling the target task to acquire optimized knowledge and avoiding unnecessarily repeated searches. Consequently, TOHO significantly improves its convergence rate of HDDPP instances.

C. Real-World Instance Test

To verify the practicability of TOHO in the real-world distribution scenario, 80 customer nodes in Changsha, Hunan Province, China are adopted as distribution nodes for the real-world instance according to [52], which is depicted in Fig. 16,

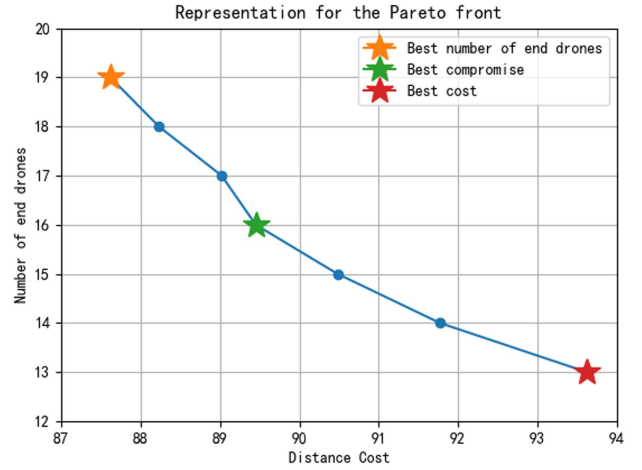


Fig. 17. The Pareto frontier of real-world instance includes 80 customer nodes in Changsha, China.

and it is extended to HDDPP real-world instance. In this experiment, the longitude and latitude coordinates of customer nodes in the real-world instance are converted into x and y coordinates in the corresponding Miller coordinate system.

To show the performance of TOHO for HDDPP routing problem with multi-objective, the Pareto frontier in the real-world instance is presented in Fig. 17, where the horizontal coordinate is distance cost, and the vertical coordinate is the number of end drones. The best compromise solution ($B-Cp$), the best distance cost solution ($B-Co$), and the best number of end drones solution ($B-Ns$) of TOHO are marked in red, orange, and green, respectively. The best compromise solution ($B-Cp$) is selected according to the knee point of the Pareto frontier, it has a relatively lower distance cost and fewer number of end drones can be recommended to decision-makers.

From Fig. 17, the distance cost and the number of end drones of the $B-Ns$ solution are 93.59 and 13 respectively. The distance cost and the number of end drones of the $B-Co$ solution are 87.62 and 19, respectively. While the $B-Cp$ solution presents a more optimized routing scheme for heterogeneous drones. Notably, the distance cost of the $B-Cp$ solution is 89.45, representing a reduction compared to the combined distance cost of $B-Ns$ and $B-Co$. Furthermore, the $B-Cp$ solution involves 16 end drones, which is fewer than the 16 end drones in the $B-Ns$ and $B-Co$ solution. The rationale behind these observations can be deduced as follows: 1) The devised topology alignment strategy, expounded in the proposed TOHO, facilitates the rapid acquisition of valuable knowledge by new instances. This, in turn, guides the search direction toward the generation of solutions of higher quality. 2) The formulated voting mechanism in the proposed TOHO, effectively selects a Pareto frontier characterized by superior diversity and convergence. Consequently, we recommend the $B-Cp$ solution to decision-makers due to its comparatively lower distance cost and reduced number of end drones.

To investigate the convergence rate of baselines on the real-world instance, the convergence curve is given in Fig. 18. From

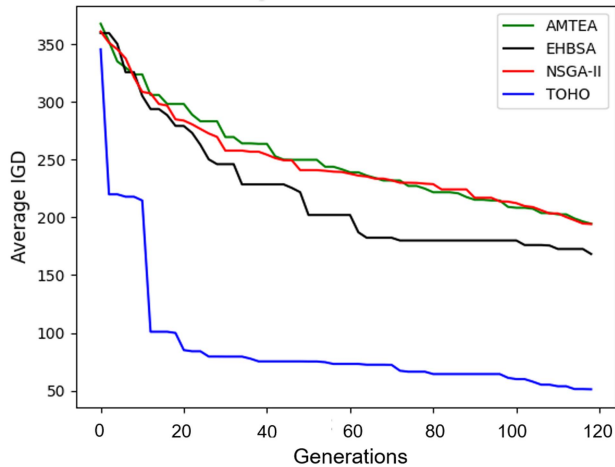


Fig. 18. The convergence curves of baselines on the real-world instance including 80 customer nodes in Changsha, China.

TABLE III
BEST-COST, BEST-NUMBER OF DRONES, AND BEST-COMPROMISE OF THE BASELINES

	NSGAII		EHBSA		AMTEA		TOHO	
	Cost	Nums	Cost	Nums	Cost	Nums	Cost	Nums
<i>B-Co</i>	98.29	20	93.25	19	90.08	20	85.17	19
<i>B-Ns</i>	112.52	12	104.96	13	101.26	13	98.61	13
<i>B-Cp</i>	105.74	17	98.67	16	93.54	16	89.39	15

Fig. 18, the proposed TOHO converges rapidly in 100 generations. Compared with NSGAII, EHBSA, and AMTEA, TOHO converges fastest and achieves better *IGD* value. The experimental results illustrate that the transfer optimization mechanism in TOHO can effectively learn the empirical knowledge from the source models, and the designed topological alignment strategy can effectively promote the positive transfer of knowledge, thus significantly speeding up the convergence speed in real-world instance.

Additionally, to further analyze the relationship between distance cost and customer satisfaction in the real-world instance, the best cost solution (*B-Co*), the best compromise solution (*B-Cp*), and the best number of end drones solution (*B-Ns*) of the baselines are compared, as shown in Table III.

As can be seen from Table III, in the best compromise, the distance cost of TOHO is 17.46%, 10.38%, and 4.64% lower than that of NSGAII, EHBSA and AMTEA, respectively. The reasons can be inferred as follows: Firstly, a clustering strategy is proposed in the first stage of TOHO, which decreases the complexity of HDDPP. Secondly, the transfer optimizer with a topological alignment strategy obtains high-quality solutions from past experience, speeds up the convergence, and gains more promising solutions. Thirdly, an ensemble selection strategy based on voting-mechanism is proposed in TOHO, which increases the selection pressure and obtains the Pareto frontier with better convergence and diversity.

TABLE IV
SENSITIVITY ANALYSIS OF PAYLOAD OF END DRONES

End drone payload	8.0	8.5	9.0	9.5	10.0	10.5	11.0	11.5	12.0
Total cost	104.98	89.37	79.21	72.74	71.20	71.08	75.12	80.34	84.55
End drone cost	22.40	25.39	28.94	32.05	34.71	35.29	39.43	44.91	49.28
Mother drone cost	82.50	63.98	50.30	40.69	36.44	35.71	35.63	35.45	35.27
Number of drones	19	18	18	17	16	15	15	14	13

D. Sensitivity Analysis

We further analyze the sensitivity of the proposed model on the end drones, i.e., the payload of small drones. For the payload of the end drone in the range of {8–12} kg, the distance cost of the mother drone and the end drones, and the number of the mounted end drones are calculated. The experimental results of sensitivity analysis of the payload of the end drone are recorded in Table IV.

Table IV illustrates a discernible trend wherein an augmentation in the payload of the end drone, from 8 kg to 10 kg, results in a noteworthy reduction in the overall distance cost. Concurrently, there is a notable increase in the distance cost specific to the end drone, accompanied by a reduction in the quantity of end drones. This observation implies that optimizing the efficiency of end-drone utilization and delivery tasks can ameliorate the challenges posed by inefficient trajectories of the mother drone.

Furthermore, the heightened payload capacity of the end drone increases the likelihood of accomplishing parcel deliveries within a sub-region in a single flight. Notably, the total distance cost is minimized when the payload reaches 10.5 kg. Beyond this threshold, as the effective payload of the end drone exceeds 10.53 kg, there is a discernible ascent in the total distance cost. Consequently, a judicious choice is to set the effective payload of the end drone at 10.53 kg to attain a low total distance cost and a relatively smaller number of mounted end drones.

VI. CONCLUSION

This research endeavors to scrutinize the intricacies of the logistics routing problem, entailing heterogeneous multi-drones engaged in pickup and delivery problems. A comprehensive optimization mathematical model is formulated to delineate an open two-echelon routing problem. To adeptly address the complexities of this routing problem, a transfer optimization with heuristic operators method is proposed. This approach integrates a topological alignment strategy, meticulously designed based on optimized source tasks, to harness the knowledge derived from these source tasks. This assimilation of knowledge is then strategically infused into the target tasks, thereby enhancing the optimization's convergence rate. Furthermore, an ensemble genetic optimization strategy employing heuristic operators according to the characteristics of HDDPP is designed to select a more promising Pareto frontier with better convergence and diversity. The efficacy of the proposed method is assessed through extensive experimentation conducted on three large-scale instances. The experimental outcomes demonstrate that the proposed transfer optimizer expedites the optimization convergence rate when juxtaposed with alternative baselines.

In future research, we aim to delve deeper into the knowledge transfer optimizer across various problem types. Additionally,

the logistics distribution problem of multi-drone in a dynamic environment is a pertinent research topic. How to design an efficient transfer optimizer to quickly generate a new routing scheme so as to adapt to the new customer demand changes is a meaningful research work that holds substantial economic and scientific significance.

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